

NET-COM

Remote automation management

Operating instructions and warnings

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SYMBOLS

The following symbols are used within this manual to indicate potential hazards.

	Important safety notice. Failure to follow these instructions may result in serious injury or damage to property. Failure to do so may result in product malfunction and create a hazardous situation.
	Important safety notice. Contact with live parts can result in death or serious injury.
	Important information for installing, programming or commissioning the product.

1 PRODUCT DESCRIPTION

NET-COM is a device for local and remote control of control units installed on automation systems for gates, garage doors and barrier gates. **Any other use must be regarded as improper and is strictly forbidden! The manufacturer shall not be liable for damages resulting from improper use of the product that differs from that stated in this manual.**

The two relay outputs can be configured in different ways. The activation command can be given from DEA remotes, from inputs or programmed time-based commands.

The product allows for managing enabling functions of remotes with time slots, events log and remote control. These functions can be extended to a device with NETLINK technology connected to the dedicated communication port.

NET-COM can function without remote connectivity. In this mode of use, it is possible to modify the operation of the outputs and the memory of the receiver remotes in a straightforward way through the DEAINSTALLER app (requires the connection of the NET-NODE interface). Advanced variations to the programming (modification of time slots and/or time-based controls of the receiver or NETLINK device connected) can be completed exclusively from the cloud.deasystem.com website and require the connection of a NET-COM via USB to a PC connected to the Internet and with DEACONNECT software started.

2 THECNICAL DATA

Connectable NET devices	1
Memorisable remote controls	1500
Frequency	433,92 MHz
Types of compatible codings	HCS rolling code - Dip-switch - DART
Programmable outputs	2 Output 0,5A/30V ===
Programmable inputs	2 Input
Power supply	24V ~ / === $\pm 5\%$ or 12V ~ / === $\pm 5\%$
Maximum power consumption	10W (with SIM) or 1,92W (without SIM LAN-EXP)
Battery	CR 2032
2G modem frequency (MHz)	850/900/1800/1900
4G modem frequency (MHz)	LTE-TDD B34/B38/B39/B40/B41 LTE-FDD B1/B2/B3/B4/B5/B7/B8/B12/B13/B18/B19/B20/B25/B26/B28/B66 UMTS/HSPA+ B1/B2/B4/B5/B6/B8/B19 GSM/GPRS/EDGE 850/900/1800/1900
Operating temperature	-10 ÷ 50 °C
Dimensions (mm)	160 x 120 x 78,5
Protection rating	IP55

3 INSTALLATION AND ASSEMBLY

WARNING NET-COM is packaged in a cardboard box identified with a label indicating the name and the product code.

It is forbidden to disperse the packaging material into the environment and leave it within reach of children, as it may be a source of danger. It must therefore be disposed of according to the laws in force.

WARNING The device can be installed and managed only by authorised and suitably instructed personnel possessing adequate technical experience.

WARNING Prior to installing NET-COM, the coverage of the GSM signal should be verified. Avoid installing the device in underground rooms or in metal box garages.

3.1 Before installation, note the following:

NET-COM is supplied with a standard antenna for receiving 433 MHz remotes. To improve reception, we recommend installing an external antenna (item 179004). The new antenna must be positioned as high as possible and above any metal or reinforced concrete structures present in the area. If necessary, use a coaxial cable with 50 ohm impedance (for example, the low-loss RG58 cable).

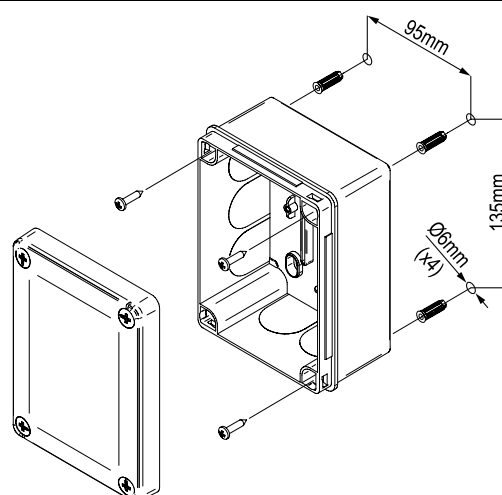
WARNING To reduce signal loss, use the shortest possible antenna cable (it must not exceed 10 m).

Install NET-COM in a zone with good GSM reception to guarantee a stable Internet connection. The product is equipped with an antenna applied under the cover which must be screwed to the appropriate connector (SMA). To improve modem reception, you can replace the supplied antenna with an external one. We recommend the GSM-LTE antenna with 2m cable (item 179009).

3.2 Proceed with the installation as described below:

Choose an assembly area that is compatible with the dimensions of the product (see **TECHNICAL DATA**) and that takes into consideration the passage of the connection cables (see **ELECTRICAL CONNECTIONS**).

- Remove the lid;
- Trace the fixing points on the wall and drill ($\varnothing 6$ mm);
- Insert the dowels;
- Center the product and fix it with the supplied screws.
- Complete the connections necessary for commissioning the product.
- Remove the insulating tab from under the coin battery.
- Screw the GSM antenna placed on the cover to the SMA connector of the circuit.
- Perform proper functioning tests.
- Close the lid.



4 ELECTRICAL CONNECTIONS



! Risk of injury and material damages due to electric shock !



! Risk of malfunction due to improper installation !

Make the connections by following the indications of the wiring diagram.

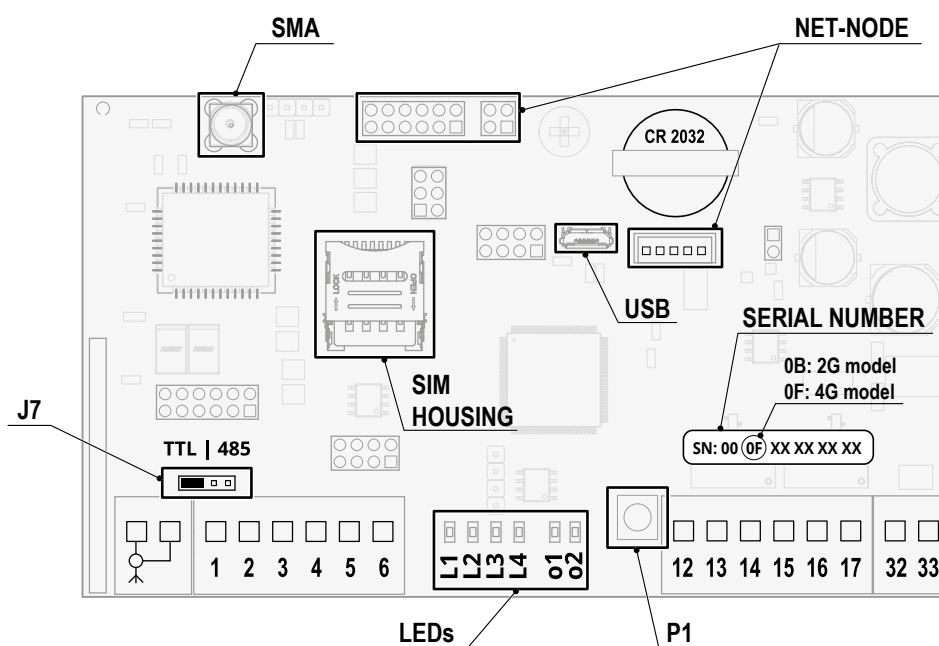
WARNING Before any connection operation, it is **MANDATORY** to disconnect the power supply.

WARNING For adequate electrical safety, all cables must have double insulation. Be sure to keep them clearly separated (**minimum 4 mm in air or 1 mm through additional insulation**) safety extra-low-voltage cables from low-voltage cables (230V ~) by arranging them within plastic raceways and securing them with appropriate clamps near the terminal blocks.

WARNING All cables should be stripped and unsheathed in the immediate vicinity of the terminals. Keep the cables slightly longer so that any excess can be removed later.

NET-COM board terminal table

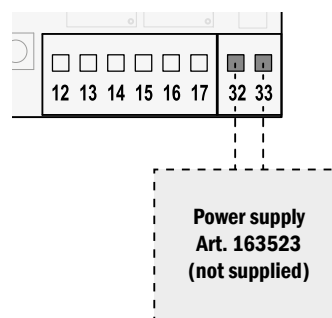
LEDs	L1, L2, L3, L4 signal the status of the device. o1, o2 signal the status of the two relay outputs.
USB	Micro USB connector for connecting and configuring NET-COM from a PC via an Internet connection.
P1	Button used to select operation via antenna, internal or external, or via LAN-EXP module. It is also used to check the quality of GSM reception and for the reset function.
SMA	Standard connector for GSM antenna supplied.
J7	Configurator of the serial port as TTL () or RS485 ().
	Remote reception antenna connection.
1, 2, 3	Serial connection of the NETLINK device controlled by NET-COM. (1) UA, (2) UB, (3) G.
4, 5, 6	Input connection. (4) common, (5) input 1, (6) input 2.
12, 13, 14	OUT 1 (12) common, (13) NO contact, (14) NC contact.
15, 16, 17	OUT 2 (15) common, (16) NO contact, (17) NC contact.
32, 33	Power supply input 24V ~ / $\pm 5\%$ or 12V ~ / $\pm 5\%$.



4.1 Power supply

The NET-COM device must be powered using a power supply suited to the consumption levels indicated in the “**TECHNICAL DATA**” paragraph.

We recommend the use of the accessory item 163523 (not supplied).

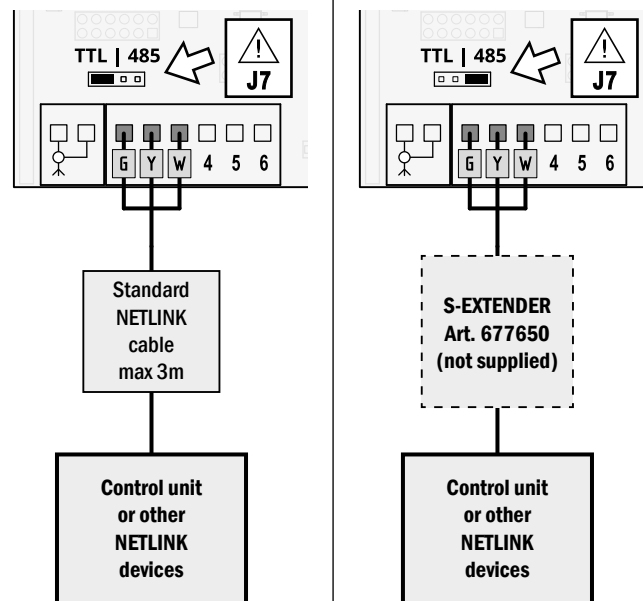


4.2 Connecting an external NETLINK device

It is possible to connect a DEA NETLINK device using the cable provided:

If NET-COM is installed at more than 3m from the device, it is necessary to use the S-EXTENDER accessory (item 677650): this signal amplifier allows for reaching a maximum distance of 50 m by using a 3x1 mm shielded cable.

In this configuration, position jumper J7 on 485.

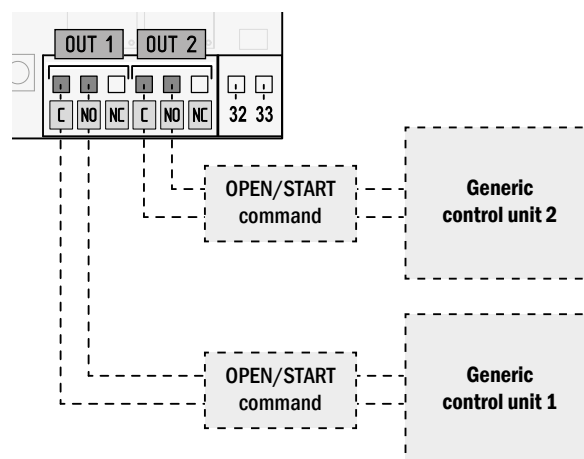


4.3 Connection of the receiver for controlling a generic automation

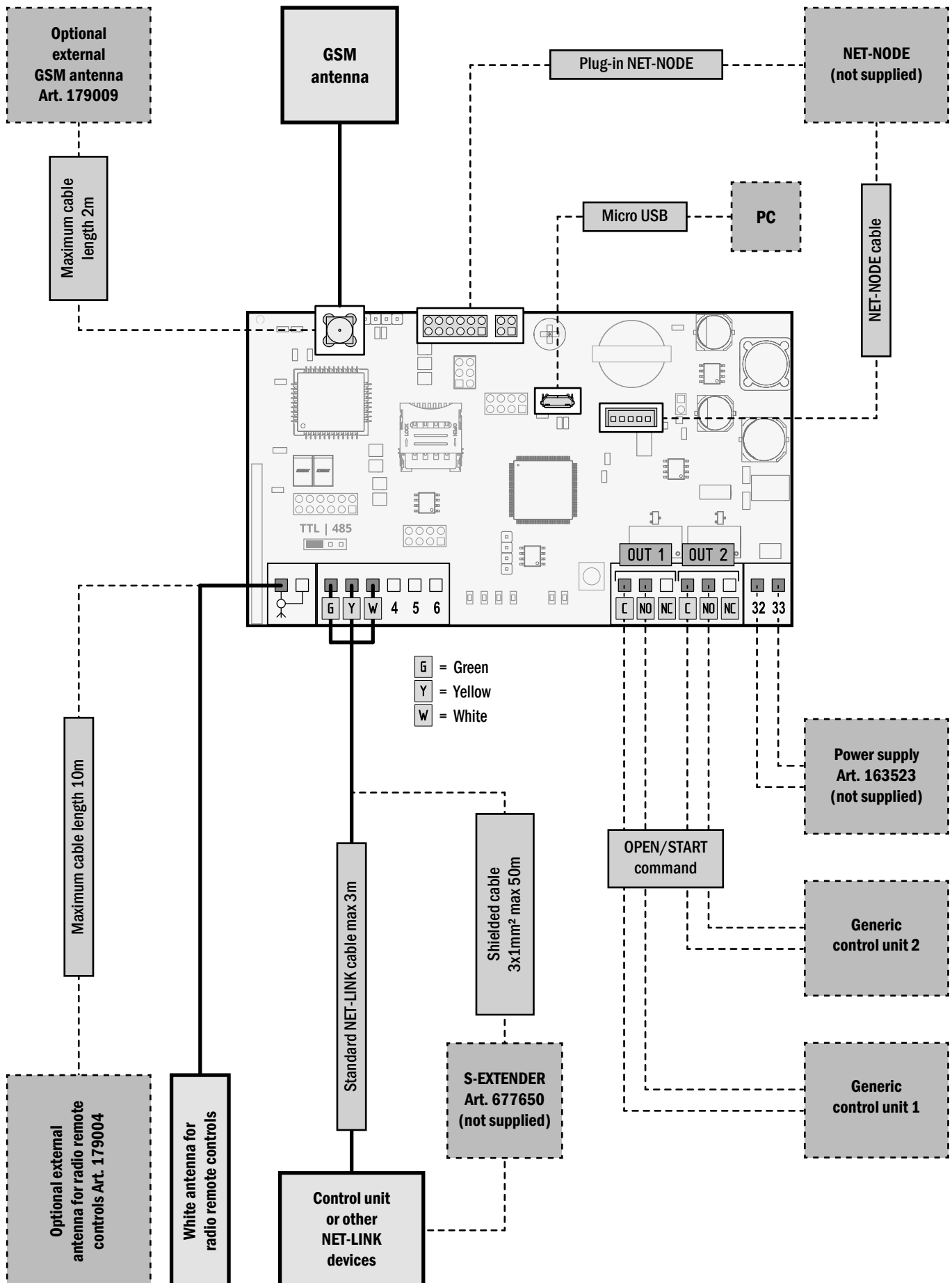
NET-COM has 2 programmable outputs;

These outputs can be activated by DEA System remotes, commands sent via APP or automated commands.

Connect the normally open contact of the desired output to the input of the control unit with **OPEN** or **START** function.



NET-COM WIRING DIAGRAM



5 CONFIGURATION

5.1 SIM configuration

NET-COM uses the data service supplied by a SIM card with **3FF format (micro SIM)**.

WARNING It is possible to insert or remove the SIM from the appropriate **"SIM Housing"** only when the device is NET-COM unpowered.

WARNING The SIM used must have the PIN disabled.

WARNING We recommend the use of specific SIM cards for IoT/M2M environments.

To enable the remote connection of NET-COM it is necessary to correctly configure the APN value of the telephone provider used. This value is factory set to "DEFAULT" to allow for the operation of the DEA SIM (included in certain versions of the product). If other SIM cards are used, the APN configuration must be performed via SMS or via CLOUD.

5.2 APN configuration via SMS

Power NET-COM with the SIM inserted and wait two minutes.

Text a message from your phone and send it to the SIM number consisting of the text

setapn:

followed by, without any space, the APN supplied to you by the telephone provider of the SIM card used.

When the command is received, NET-COM replies with an SMS confirmation message and restarts with the new settings.

5.3 APN configuration via CLOUD

Connect NET-COM to the PC via the USB cable, making sure you have an active Internet connection and the DEAcconnect software* (Windows only) connected to the GATEWAY.

Connect to the website: cloud.deasystem.com

* **DEAcconnect is available for downloading under DOWNLOAD > Software**

Perform the registration and then navigate to the **"Tools NET-BOX"** section. Enter the **"Gateway info"** tab and input the serial number of the NET-COM gateway in the designated field (the serial number can be seen in the "gateway (NET-BOX, NET-COM)" window of the DEAcconnect software).

By clicking on **"gateway connection"**, the connection is established, and the APN field becomes active.

Modify the field by entering the desired APN, confirm by clicking on **"Change SIM Settings"**, and wait for the NET-COM device to reboot.

To verify the correct functionality of the new APN, it is necessary to disconnect the USB cable from the PC, connect the 12/24V power supply, and wait for NET-COM to establish a connection.

5.4 GSM antenna configuration and signal verification

NET-COM is supplied with an antenna with cable and SMA connector applied under the box lid.

If it is necessary to install the antenna further away (for example to improve modem reception), it is recommended to use the "ANTENNA GSM-LTE" (item 179009 not provided) external antenna with a length of up to 2m to be connected to the dedicated SMA connector.

Attention: The use of external antennas other than the model indicated does not guarantee the correct functioning of the product and could cause malfunctions.

it is possible to configure NET-COM for connection via the accessory "LAN-EXP LAN module for NET-COM" (accessory not included).

Power the NET-COM device and press and hold **P1** for 5s to visualise the signal of the type of antenna configured:

- **GSM antenna** -> L1+L4 or L2+L3 flashing
- **LAN-EXP module** -> L1+L3 flashing

The signaling described lasts 5s. During this time it is possible to press **P1** to change the desired configuration (let the 5s elapse to confirm the selection).

When the type of signal set is changed, an automatic restart of the NET-COM will follow.

5.5 Signal strength check

The strength of the telephone provider's signal can only be checked when NET-COM is connected to the internet (LED L1 on steady).

Press and release button **P1** to start communication. The LEDs will display the signal level for 10s as indicated in the following table.

L1	L2	L3	L4	Signal level
•	•	•	•	Maximum
•	•	•		Good
•	•			Sufficient
•				Poor

To highlight that leds L1-L4 are signaling the level of the modem reception signal, during this signal led **o1** and **o2** flash rapidly. Press and release button **P1** to end the signaling immediately.

5.6 Programming

The data of the NET-COM receiver and of the connected NETLINK device can be configured using the Web interface **cloud.deasystem.com**.

To complete the data synchronisation, NET-COM must be connected to the Internet.

In the absence of connectivity, it is possible to complete the synchronisation by connecting NET-COM via USB to the PC. Start the DEAcconnect software (which can be downloaded from the DEAccloud download area) from the latter and complete the operation from **cloud.deasystem.com**.

WARNING The commissioning of a NETLINK control unit must always be carried out locally through manual programming or from the DEAcinstaller app before connecting the NET-COM gateway.

Connecting NET-NODE to NET-COM it is possible to complete the programming of the sole parameters of the remotes of the receiver using the DEAcinstaller app.

WARNING To configure the following functions, it is necessary to use DEAccloud (**cloud.deasystem.com**):

- Enabling of the remotes at specific time slots.
- Activation of the commands at specific time slots.
- Activation of the commands from telephone call.

5.7 RESET procedure

The reset procedure can be distinguished into two modes: “**soft**” and “**hard**” reset.

The “soft” reset procedure will involve the following steps:

- Deletion of stored remote controls on NET-COM.
- Restoration of NET-COM configuration parameters to default values.
- Removal of configured time-based commands and/or time slots.

“Soft” reset procedure:

- With NET-COM powered off, press and hold **P1** button.
- Power on the board while keeping the button pressed.
- Wait for the LED indicators L1, L2, L3, and L4 to turn on (approximately 10 seconds).
- Release the **P1** button.

The “hard” reset procedure will involve the following steps:

- Restoration of NET-COM to factory settings.
- Removal of the configured APN (Access Point Name).
- Deletion of the installer account associated with NET-COM.

“Hard” reset procedure:

- With NET-COM powered off, press and hold **P1** button.
- Power on the board while keeping the button pressed.
- Wait for the LED indicators L1, L2, L3, and L4 to turn on and then turn off (approximately 20 seconds).
- Release the **P1** button.

6 SIGNALLING AND DIAGNOSTICS LED

6.1 Status of outputs

LEDs **o1** and **o2** lit (red) signal that the relative output is active.

When LEDs **o1** and **o2** flash rapidly, it means that the modem signal strength visualisation is under way. Refer to the “**GSM antenna configuration and signal verification**” paragraph.

6.2 Internet connection

STATE 1 – Not connected

L1 flashes once every 10 s when NET-COM is not connected to the Internet.

This status is permanent when an active SIM is not inserted or when the PIN was not disabled.

STATE 2 – Scanning of operators or connection under way

L1 flashes once rapidly when NET-COM is scanning the available operators or is connecting to one of these.

WARNING The operator scanning process and connection may take up to 10 minutes.

STATE 3 – Connection to the Internet

L1 is lit steady when NET-COM is connected to the Internet.

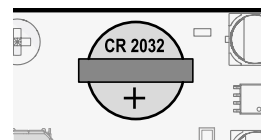
If the “STATE 1” and “STATE 2” signals alternate constantly, it means that NET-COM is unable to complete the connection to the Internet. A few possible causes may be:

- The SIM is active and with the PIN disabled but the data service is unavailable due to lack of credit.
- The telephone provider supported by the SIM do not have sufficient signal strength. It is possible to try and use the external antenna.

6.3 Continuous loss of the clock settings

The CR2032 inserted in the NET-COM device ensures the correct operation of the clock in the absence of power for 7 consecutive days. When there is an Internet connection, the clock is continuously verified and synchronised on the basis of the settings programmed with DEAccloud.

In case of continuous loss of time, we recommend replacing the battery.



WARNING: The use of batteries of the wrong type can cause a risk of fire and/or explosion of the product. Use only batteries of the specified type.

7 PRODUCT DISPOSAL

DISASSEMBLY

The dismantling of the product must be carried out by qualified personnel in accordance with prevention and safety and with reference to the assembly instructions but in reverse order. Before starting the disassembly, disconnect the power supply and protect against a possible reconnection.

DISPOSAL

Disposal of the product must be performed in accordance with national and local disposal regulations. The product (or its individual parts) must not be disposed of with other household waste.



WARNING In compliance with EU Directive 2012/19/EU on waste electrical and electronic equipment (WEEE), this electrical product should not be treated as municipal mixed waste. Please dispose of the product and bring it to the collection for an appropriate local municipal recycling.



Attention!

This product contains a “button” type battery. Keep the new and used batteries out of reach of children.

! DO NOT INGEST THE BATTERY ! If the battery is ingested, it may cause serious internal burns in just 2 hours and bring to death.

In case of suspected ingestion or insertion in other body parts, immediately contact a doctor.



RÉCAPITULATIF DES AVERTISSEMENTS

ATTENTION! IMPORTANTES INSTRUCTIONS DE SÉCURITÉ. LIRE ET SUIVRE ATTENTIVEMENT TOUTES LES MISES EN GARDE ET LES INSTRUCTIONS QUI ACCOMPAGNENT LE PRODUIT CAR UNE INSTALLATION ERRONÉE PEUT PROVOQUER DES DOMMAGES AUX PERSONNES, ANIMAUX OU CHOSSES. LES MISES EN GARDE ET LES INSTRUCTIONS FOURNISSENT D'IMPORTANTES INDICATIONS AU SUJET DE LA SÉCURITÉ, L'INSTALLATION, L'UTILISATION ET LA MAINTENANCE. CONSERVER LES INSTRUCTIONS POUR LES JOINDRE AU DOSSIER TECHNIQUE ET POUR DE FUTURES CONSULTATIONS.

■ **ATTENTION** Ne pas laisser les enfants jouer avec l'appareil. L'appareil peut être utilisé par des enfants âgés d'au moins 8 ans, par des personnes souffrant d'une déficience physique, mentale ou sensorielle réduite, ou en général par toute personne sans expérience ou, en tout cas, avec l'expérience requise, à condition que l'appareil soit utilisé sous surveillance ou que les utilisateurs aient reçu une formation adéquate sur l'utilisation sûre de l'appareil et soient conscients des dangers liés à son utilisation. ■ **ATTENTION** Les commandes

à installation fixe (boutons, etc.) doivent être situées hors de la portée des enfants à au moins 150 cm de hauteur du sol. Ne pas permettre aux enfants de jouer avec l'appareil, les commandes fixes ou avec les radiocommandes de l'installation. ■ **ATTENTION** L'utilisation du produit dans des conditions

anormales, non autorisées par le fabricant, peut entraîner des situations de danger ; respecter les conditions prévues sur cette notice d'utilisation. ■

ATTENTION DEA System vous rappelle que le choix, la position et l'installation de tous les dispositifs et les matériaux qui constituent l'ensemble complet de la fermeture, doivent être exécutés conformément aux Directives Européennes 2006/42/CE (Directive Machines) et ses modifications ultérieures, 2014/53/UE (Directive RED). Dans tous pays extracommunautaires, non seulement vous devez suivre les normes spécifiques en vigueur mais, pour atteindre un niveau de sûreté suffisant, on vous conseille d'observer aussi les prescriptions des Directives susmentionnées. ■ **ATTENTION** N'utiliser en aucun cas l'appareil

en présence d'une atmosphère explosive ou dans des environnements qui peuvent être agressifs et endommager des parties du produit. Vérifier que les températures dans le lieu d'installation soient appropriées et respectent les températures déclarées sur l'étiquette du produit. ■ **ATTENTION** Quand on

opère avec la commande à « action maintenue », s'assurer que personne ne se trouve dans la zone de manutention de l'automatisme. ■ **ATTENTION** Vérifier

qu'en amont du réseau d'alimentation de l'installation, il y ait un interrupteur ou un disjoncteur magnétothermique omnipolaire qui permette la déconnexion complète dans les conditions de la catégorie de la surtension III. ■ **ATTENTION**

Afin d'assurer une sécurité électrique, gardez toujours nettement séparés (minimum 4 mm en air ou 1 mm à travers l'isolation) le câble d'alimentation 230V des câbles à très basse tension de sécurité (alimentation des moteurs, commandes, électro-serrure, antenne, alimentation des circuits auxiliaires)

éventuellement en les fixant à l'aide de pattes d'attache appropriées à proximité des bornes. ■ **ATTENTION** Si le câble d'alimentation est endommagé, il doit être remplacé par le constructeur ou par son service d'assistance technique ou, quoi qu'il en soit, par une personne possédant une qualification similaire, de manière à empêcher tous les risques. ■ **ATTENTION** Toute opération d'installation, de maintenance, de nettoyage ou de réparation de toute l'installation doit être exécutée exclusivement par du personnel qualifié. Opérez toujours quand l'alimentation est coupée, et conformez-vous rigoureusement à toutes les normes en matière d'installations électriques en vigueur dans le pays où cette automatisation doit être installée. Le nettoyage et la maintenance destinée à être effectués par l'utilisateur ne doivent pas être accomplis par des enfants sans surveillance. ■ **ATTENTION** L'utilisation de pièces de rechange non indiquées par DEA System et/ou un réassemblage incorrect peuvent être potentiellement dangereux pour les personnes, les animaux et les choses. De plus, cela peut provoquer des dysfonctionnements du produit. Par conséquent, utilisez toujours les pièces indiquées par DEA System et suivez les instructions données pour l'assemblage. ■ **ATTENTION** Après le réglage, le respect des valeurs limites réglementaires doit être détecté à l'aide d'un instrument de mesure d'impact de force. La sensibilité de la détection d'obstacle peut être ajustée progressivement à la porte (voir les instructions de programmation). Le fonctionnement du dispositif anti-écrasement doit être vérifié après chaque réglage manuel. La modification manuelle de la force ne peut être effectuée que par du personnel qualifié en effectuant le test de mesure selon la norme EN 12445. Les modifications du réglage de la force doivent être documentées dans le manuel de la machine. ■ **ATTENTION** La conformité aux exigences de la norme EN 12453 du dispositif de détection d'obstacles interne est garantie seulement si utilisé en conjonction avec des moteurs équipés d'encodeurs. ■ **ATTENTION** Tout dispositif de sécurité externe éventuellement utilisé afin de respecter les limites des forces d'impact doit être conformes à la norme EN 12978. ■  **ATTENTION** Conformément à la Directive 2012/19/EG sur les déchets d'équipements électriques et électroniques (WEEE), ce produit électrique ne doit en aucun cas être mis au rebut sous forme de déchet municipal non trié. Veuillez vous débarrasser de ce produit en le renvoyant au point de ramassage local dans votre municipalité, à des fins de recyclage.

TOUT CE QUI N'EST PAS PRÉVU EXPRESSÉMENT DANS LE MANUEL D'INSTALLATION, EST INTERDIT. LE BON FONCTIONNEMENT DE L'OPÉRATEUR EST GARANTI UNIQUEMENT SI LES DONNÉES MENTIONNÉES SONT RESPECTÉES. LA FIRME NE RÉPOND PAS DES DOMMAGES CAUSÉS PAR LE NON-RESPECT DES INDICATIONS MENTIONNÉES DANS CE MANUEL. EN LAISSANT INALTÉRÉES LES CARACTÉRISTIQUES ESSENTIELLES DU PRODUIT, DEA SYSTEM SE RÉSERVE LE DROIT D'APPORTER À TOUT MOMENT LES MODIFICATIONS QU'ELLE RETIENT IMPORTANTES POUR AMÉLIORER SUR LE CARACTÈRE TECHNIQUE, DE CONSTRUCTION ET COMMERCIAL LE PRODUIT, SANS S'ENGAGER À METTRE À JOUR LA PRÉSENTE PUBLICATION.

